

§2.5

$$y'' + 3y = t^3 - 1$$

P1. Solution: $\chi(t) = A_0 + A_1 t + A_2 t^2 + A_3 t^3$

$$\begin{aligned} L[\chi](t) &= \chi''(t) + 3\chi(t) \\ &= [2A_2 + 6A_3 t] + 3[A_0 + A_1 t + A_2 t^2 + A_3 t^3] \\ &= (3A_0 + 2A_2) + (3A_1 + 6A_3)t \\ &\quad + 3A_2 t^2 + 3A_3 t^3 \end{aligned}$$

$$\Rightarrow \begin{cases} 3A_3 = 1 \\ 3A_2 = 0 \\ 3A_1 + 6A_3 = 0 \\ 3A_0 + 2A_2 = -1 \end{cases} \Rightarrow \begin{cases} A_3 = \frac{1}{3} \\ A_2 = 0 \\ A_1 = -\frac{2}{3} \\ A_0 = -\frac{1}{3} \end{cases}$$

Thus, $\chi(t) = \frac{t^3}{3} - \frac{2}{3}t - \frac{1}{3}$

P2. Solution: $y'' + 4y' + 4y = t e^{\alpha t}$

Characteristic equation: $r^2 + 4r + 4 = 0$
 $\Rightarrow r = -2$

Case i) $\alpha \neq -2$. Then

$$\chi(t) = e^{\alpha t} (A_0 + A_1 t)$$

$$\begin{aligned} L[\chi](t) &= \chi''(t) + 4\chi'(t) + 4\chi(t) \\ &= e^{\alpha t} [\alpha^2 A_0 + 2\alpha A_1 + \alpha^2 A_1 t] \\ &\quad + 4e^{\alpha t} [\alpha A_0 + A_1 + \alpha A_1 t] \\ &\quad + 4e^{\alpha t} [A_0 + A_1 t] \\ &= e^{\alpha t} [(\alpha^2 + 4\alpha + 4)A_1 t \\ &\quad + (\alpha^2 + 4\alpha + 4)A_0 + (2\alpha + 4)A_1] \\ &\Rightarrow (\alpha + 2)^2 A_1 = 1 \end{aligned}$$

$$(\alpha + 2)^2 A_0 + 2(\alpha + 2)A_1 = 0 \Rightarrow (\alpha + 2)A_0 + 2A_1 = 0$$

$$\Rightarrow A_1 = \frac{1}{(\alpha + 2)^2}$$

$$A_0 = \frac{-2}{(\alpha + 2)^3}$$

$$\Rightarrow \chi(t) = e^{\alpha t} \left(\frac{t}{(\alpha + 2)^2} - \frac{2}{(\alpha + 2)^3} \right)$$

Case ii) $\alpha = -2$. Then.

$$\begin{aligned} \chi(t) &= e^{-2t} t^2 (A_0 + A_1 t) \\ &= e^{-2t} v(t) \end{aligned}$$

$$\Rightarrow L[\chi](t) = e^{-2t} v''(t)$$

$$\begin{aligned} \Rightarrow v''(t) &= t \\ \Rightarrow v'(t) &= \frac{t^2}{2} \\ \Rightarrow v(t) &= \frac{t^3}{6} \end{aligned}$$

$$\Rightarrow \chi(t) = \frac{t^3 e^{-2t}}{6}$$

P3. Solution: $y'' - y = t^2 e^t$

Characteristic equation: $r^2 - 1 = 0$
 $r = \pm 1$

Thus, $\chi(t) = e^t (A_0 + A_1 t + A_2 t^2)$

$$\begin{aligned} L[\chi](t) &= \chi''(t) - \chi(t) \\ &= e^t [2(A_0 + A_1) + (A_0 + 4A_1 + 6A_2)t \\ &\quad + (A_1 + 6A_2)t^2 + A_2 t^3] \\ &\quad - e^t [A_0 + A_1 t + A_2 t^2] \\ &= e^t [2(A_0 + A_1) + (4A_1 + 6A_2)t + 6A_2 t^2] \end{aligned}$$

$$\Rightarrow \begin{cases} 6A_2 = 1 \\ 4A_1 + 6A_2 = 0 \\ 2(A_0 + A_1) = 0 \end{cases} \Rightarrow \begin{cases} A_2 = \frac{1}{6} \\ A_1 = -\frac{1}{4} \\ A_0 = \frac{1}{4} \end{cases}$$

$$\Rightarrow \chi(t) = t \left(\frac{t^2}{6} - \frac{t}{4} + \frac{1}{4} \right) e^t$$

P5. Solution: $y'' + 2y' + y = e^{-t}$

Characteristic equation: $r^2 + 2r + 1 = 0$
 $r = -1$

Then $\chi(t) = e^{-t} t^2 (A_0 + A_1 t)$
 $= e^{-t} v(t)$

$$\begin{aligned} \Rightarrow L[\chi](t) &= \chi''(t) + 2\chi'(t) + \chi(t) \\ &= e^{-t} v''(t) \end{aligned}$$

$$\Rightarrow v''(t) = 1$$

$$\Rightarrow v'(t) = t$$

$$\Rightarrow v(t) = \frac{t^2}{2}$$

$$\Rightarrow \underline{\chi(t) = \frac{t^2}{2} e^{-t}}$$

P.7. Solution: $y'' + 4y = t \sin 2t$

$$\sin 2t = \text{Im}\{e^{i2t}\}$$

We need to find a particular solution $\phi(t)$ for the equation

$$y'' + 4y = t e^{i2t}$$

Characteristic equation: $r^2 + 4 = 0$
 $r = \pm 2i$

$$\Rightarrow \phi(t) = e^{i2t} t(A_0 + A_1 t)$$

$$\phi'(t) = e^{i2t} (A_0 + 2(A_0 i + A_1) t + i 2 A_1 t^2)$$

$$\phi''(t) = e^{i2t} (4A_0 i + 2A_1 + (-4A_0 + i 2A_1) t - 4A_1 t^2)$$

$$\Rightarrow \mathcal{L}[\phi](t) = \phi'' + 4\phi$$

$$= e^{i2t} [(8A_1 i) t + 4A_0 i + 2A_1]$$

$$\Rightarrow 8A_1 i = 1 \quad \left\{ \begin{array}{l} A_1 = -\frac{i}{8} \\ 4A_0 i + 2A_1 = 0 \end{array} \right.$$

$$4A_0 i + 2A_1 = 0 \quad A_0 = \frac{1}{16}$$

$$\Rightarrow \phi(t) = e^{i2t} t \left(-\frac{i}{8} t + \frac{1}{16} \right)$$

$$= (\cos 2t + i \sin 2t) \left(-\frac{i}{8} t + \frac{1}{16} \right) t$$

$$= t \left[\frac{t}{8} \sin 2t + \frac{1}{16} \cos 2t + i \left(-\frac{t}{8} \cos 2t + \frac{\sin 2t}{16} \right) \right]$$

$$\Rightarrow \chi(t) = \text{Im}\{\phi(t)\}$$

$$= t \left(-\frac{t}{8} \cos 2t + \frac{\sin 2t}{16} \right)$$

$$= \underline{16t \left(-2t \cos 2t + \sin 2t \right)}$$

P.10. Solution: $y'' - 2y' + 5y = 2(\cos^2 t) e^t$

$$\cos^2 t = \frac{\cos 2t + 1}{2}$$

$$\Rightarrow y'' - 2y' + 5y = \frac{e^t \cos 2t + e^t}{2} \quad \chi_1(t)$$

We need to find a particular solution for

$$y'' - 2y' + 5y = e^t \cos 2t$$

and $\chi_2(t)$ for

$$y'' - 2y' + 5y = e^t$$

so that $\chi(t) = \chi_1(t) + \chi_2(t)$

(i) $e^t \cos 2t = \text{Re}\{e^{(1+2i)t}\}$, $y'' - 2y' + 5y = e^{(1+2i)t}$

Characteristic equation $r^2 - 2r + 5 = 0$

$$r_1 = 1 + 2i, r_2 = 1 - 2i$$

$$\phi_1(t) = A_0 t e^{(1+2i)t}$$

$$\phi_1'(t) = A_0 [1 + (1+2i)t] e^{(1+2i)t}$$

$$\phi_1''(t) = A_0 [(-3+4i)t + (2+4i)] e^{(1+2i)t}$$

$$\Rightarrow \mathcal{L}[\phi_1](t) = \phi_1''(t) - 2\phi_1'(t) + 5\phi_1(t)$$

$$= (4A_0 i) e^{(1+2i)t}$$

$$\Rightarrow 4A_0 i = 1 \Rightarrow A_0 = -\frac{i}{4}$$

$$\Rightarrow \phi_1(t) = -\frac{i}{4} e^{(1+2i)t}$$

$$= \frac{t}{4} e^t [\sin 2t - i \cos 2t]$$

$$\Rightarrow \chi_1(t) = \text{Re}\{\phi_1(t)\}$$

$$= \frac{t}{4} e^t \sin 2t$$

(ii) $y'' - 2y' + 5y = e^t$

$$\chi_2(t) = A_0 e^t$$

$$\Rightarrow \mathcal{L}[\chi_2](t) = (A_0 - 2A_0 + 5A_0) e^t$$

$$= 4A_0 e^t$$

$$\Rightarrow 4A_0 = 1 \Rightarrow A_0 = \frac{1}{4}$$

$$\Rightarrow \chi_2(t) = \frac{e^t}{4}$$

Thus, the particular solution is given by

$$\chi(t) = \chi_1(t) + \chi_2(t)$$

$$= \frac{t}{4} e^t \sin 2t + \frac{e^t}{4}$$

$$= \underline{\frac{e^t}{4} (t \sin 2t + 1)}$$

P11. Solution: $y'' + y' - 6y = \sin t + te^{2t}$.

i) We first need to find a particular solution ^{for}

$$y'' + y' - 6y = \sin t = \text{Im}\{e^{it}\}$$

$$r^2 + r - 6 = 0$$

$$\Rightarrow r_1 = 2, r_2 = -3$$

Consider the equation

$$y'' + y' - 6y = e^{it}$$

$$\phi_1(t) = A_0 e^{it}$$

$$\Rightarrow L[\phi_1](t) = (-A_0 + A_0 i - 6A_0) e^{it} = (-7 + i) A_0 e^{it}$$

$$\Rightarrow (-7 + i) A_0 = 1 \Rightarrow A_0 = \frac{1}{-7 + i}$$

$$\begin{aligned} \Rightarrow \chi_1(t) &= \text{Im}\{\phi_1(t)\} \\ &= \text{Im}\left\{\frac{1}{-7 + i} e^{it}\right\} \\ &= \frac{1}{50} (-6\sin t - 7\cos t) \end{aligned}$$

ii) We then find a particular solution $\chi_2(t)$ for

$$y'' + y' - 6y = te^{2t}$$

$$\chi_2(t) = t(A_0 + A_1 t) e^{2t}$$

$$\begin{aligned} L[\chi_2](t) &= \chi_2''(t) + \chi_2'(t) - 6\chi_2(t) \\ &= (5A_0 + 2A_1 + 10A_1 t) e^{2t} \end{aligned}$$

$$\Rightarrow \begin{cases} 10A_1 = 1 \\ 5A_0 + 2A_1 = 0 \end{cases} \Rightarrow \begin{cases} A_1 = \frac{1}{10} \\ A_0 = -\frac{1}{25} \end{cases}$$

$$\Rightarrow \chi_2(t) = t\left(\frac{1}{10} - \frac{t}{25}\right) e^{2t}$$

Thus,

$$\chi(t) = \chi_1(t) + \chi_2(t)$$

$$= \frac{1}{50} (-6\sin t - 7\cos t) + t\left(\frac{1}{10} - \frac{t}{25}\right) e^{2t}$$

P15. Solution: $y'' + y = \cos t \cos 2t$

$$\cos t \cos 2t = \frac{1}{2} \cos 3t + \frac{1}{2} \cos t$$

$$y'' + y = \frac{1}{2} \cos 3t + \frac{1}{2} \cos t$$

i) We first need to find a particular solution $\chi_1(t)$ for $y'' + y = \frac{1}{2} \cos 3t$. Since $\cos 3t = \text{Re}\{e^{i3t}\}$ let us consider the equation $y'' + y = \frac{1}{2} e^{i3t}$.

$$r^2 + 1 = 0 \Rightarrow r = \pm i$$

$$\phi_1(t) = A_0 e^{i3t}$$

$$\begin{aligned} L[\phi_1](t) &= \phi_1''(t) + \phi_1(t) \\ &= (-9A_0 + A_0) e^{i3t} \\ &= -8A_0 e^{i3t} \end{aligned}$$

$$\Rightarrow -8A_0 = \frac{1}{2} \Rightarrow A_0 = -\frac{1}{16}$$

$$\Rightarrow \phi_1(t) = -\frac{1}{16} e^{i3t}$$

$$\Rightarrow \chi_1(t) = \text{Re}\{\phi_1(t)\} = -\frac{1}{16} \cos 3t$$

ii) We then need to find a particular solution.

$\chi_2(t)$ for $y'' + y = \frac{1}{2} \cos t$. Since $\cos t = \text{Re}\{e^{it}\}$ let us consider the equation $y'' + y = \frac{1}{2} e^{it}$

$$\phi_2(t) = A_0 t e^{it}$$

$$\begin{aligned} L[\phi_2](t) &= \phi_2''(t) + \phi_2(t) \\ &= [-A_0 t + 2A_0 i + A_0 t] e^{it} \\ &= 2A_0 i e^{it} \end{aligned}$$

$$\Rightarrow 2A_0 i = \frac{1}{2} \Rightarrow A_0 = -\frac{i}{4}$$

$$\Rightarrow \phi_2(t) = -\frac{i}{4} t e^{it}$$

$$\begin{aligned} \Rightarrow \chi_2(t) &= \text{Re}\{\phi_2(t)\} \\ &= \text{Re}\left\{-\frac{it}{4} (\cos t + i \sin t)\right\} \\ &= \text{Re}\left\{\frac{t}{4} \sin t - \frac{i}{4} \cos t\right\} \\ &= \frac{t}{4} \sin t \end{aligned}$$

Thus

$$\begin{aligned} \chi(t) &= \chi_1(t) + \chi_2(t) \\ &= -\frac{1}{16} \cos 3t + \frac{t}{4} \sin t \end{aligned}$$